# Group

- **Set + one operation** (usually + or ×)
- **Requirements:**
    1. **Closure:** $a*b \in G$
    2. **Associativity:** $(ab)c = a(bc)$
    3. **Identity:** $\exists$ e such that $ea = ae = a$
    4. **Inverses:** $\forall a, \exists a^{-1}$ such that $a*a^{-1} = e$
- **Examples:** $(\mathbb{Z}, +)$, $(\mathbb{R}\{0\}, \times)$

# Ring

- **Set + two operations** (+, ×)
- **Requirements:**
    1. **(G, +) is abelian group** (commutative)
    2. **× is associative**
    3. **Distributive:** $a\times(b+c) = a\times b + a\times c$
- **May have:** × identity (1), × commutativity
- **Examples:** $\mathbb{Z}$, matrices, polynomials

# Field

- **Set + two operations** (+, ×)
- **Requirements:**
    1. **(F, +) is abelian group**
    2. **(F{0}, ×) is abelian group**
    3. **Distributive laws hold**
- **Examples:** $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}_p$ (p prime)

# Vector Space

- **Set V over field K**
- **Requirements:**
    1. **(V, +) is abelian group**

2. **Scalar multiplication:** $K \times V \to V$
3. **8 axioms** from your PDF

- **Key:** Can add vectors, multiply by scalars
- **Examples:** $\mathbb{R}^n$, polynomials, matrices

# Subspace

- **Subset $W \subseteq V$** where V is vector space over K
- **Quick check:**
  1. **$0 \in W$**
  2. **Closed under +:** $u,v \in W \Rightarrow u+v \in W$
  3. **Closed under ·:** $a \in K, u \in W \Rightarrow a \cdot u \in W$
- **Even quicker:** Check $a \cdot u + b \cdot v \in W$ for all $a,b \in K$, $u,v \in W$

# Memory Aid

- **Group:** One operation, all inverses
- **Ring:** Two operations, + has inverses, × may not
- **Field:** Two operations, both have inverses
- **Vector Space:** Vectors + scalars, geometric structure
- **Subspace:** Vector space within vector space

**Test for subspace:** "Can I do all vector operations here without leaving?"